One-page document

1. **Core Idea:** ***need/problem***

- Insufficient citizen participation in proper waste disposal and recycling.
- Inefficient recycling initiatives for positive behaviour.

We will reduce street pollution and encourage citizens, especially the younger generation, to recycle by using our application. The app is based on gamification, allowing users to earn rewards when they recycle correctly. These reward points can be exchanged for free public transportation or discounts on government-supported services, such as electric vehicle charging stations or museum visits.

2. **Digital Transformation:** ***The role and significance of digital transformation***

There is a need for an integration of a smart digital ecosystem to combine QR codes, artificial intelligence, cloud technology, and a mobile application. Instead of simply throwing away recyclable waste, users scan a QR code on the smart bin, allowing the system to securely identify them and record their recycling activity. AI then verifies the type of waste and checks whether it has been disposed of correctly. Once confirmed, users automatically receive reward points that can be redeemed for public transportation, bike sharing, e-scooter rides, or even museum and theater tickets. At the same time, every recycling activity is stored in the cloud, providing municipalities with real-time data on recycling rates, waste composition, and bin usage. This data enables smarter decision-making, more efficient waste collection, lower operational costs, and supports the development of more sustainable cities

3. **Green & Sustainability:** ***Green and sustainability aspects***

Motivation is one factor affecting people to participate in good behaviour. One way to increase it is with gamification and reward, raising more awareness for sustainability. Real-time tracing in the waste system, improving feasibility in garbage collecting, using people's data to analyze and improve recycling operations.

4. **Economic & Competition:** ***Economic and competition aspects***

Companies providing smart bins and waste collection optimization exist. For example, Sensoneo is a global leader in data-driven smart waste management and operational efficiency. Our idea takes this one step further to introduce gamification and community feedback to strengthen and support sustainable travel options.

5. **Data:** ***Making digital data understandable***

We track two key data points:

- User Behaviour: who is recycling, when, and where;
- Bin Data: which includes live fill levels, material types, and contamination rates.

The user scans a QR code on our app to link their profile. Inside the bin, IoT sensors check its capacity, while an AI optical scanner instantly verifies the exact type of waste being deposited.

This data makes our entire ecosystem viable. For the citizens, AI verification prevents fraud, allowing us to instantly convert their good habits into travel rewards. For waste managers, it eliminates blind spots. Instead of driving fixed routes to empty half-full bins, cities use our live dashboard to dispatch trucks only where needed, saving fuel, cutting emissions and giving sponsors concrete analytics on their environmental impact.

6. **Business Models:** ***Potential revenue sources and business models***

Revenue sources

The public sector is our primary partner. Because increasing recycling rates and gaining accurate, real-time data on waste flows is a clear public interest, the government finances the operation and maintenance of the platform through a fixed contribution. In return, the city receives a reliable tool for tracking, reporting and improving its waste management, ensuring the long-term stability of the service.

Our second source of revenue comes from the partner companies whose rewards are offered through the platform. For every reward redeemed by a user, the company pays us a commission, as our platform brings them new, engaged customers. For example, if a user collects 1,000 points with a value of 10€, we retain a 10% commission, which the partner company covers in exchange for the acquired customer and the visibility gained within the application.

Plans for the future

As the platform grows, two further revenue streams can be introduced. The first is a share of the value generated by the recycling process itself, earned through the sale of the collected materials to recycling operators. The second is advertising space within the application, offered to sustainability-oriented brands. Together, these would diversify our income and reduce dependence on any single source.

Business model

<table>
<tr>
<td rowspan="2">Key Partners
• Municipalities & government
• Recycling operators
• Transport & mobility partners
• Technology providers (IoT, AI)</td>
<td>Key Activities
• Operating smart bins (IoT & AI sorting)
• Running the rewards app
• Collecting & reporting waste data</td>
<td rowspan="2">Value Propositions
• Citizens: free recycling rewarded with travel benefits
• Cities: real-time waste data & less pollution</td>
<td>Customer Relationships
• Gamified, self-service app
• Points & incentives
• Community feedback</td>
<td rowspan="2">Customer Segments
• Citizens (mainly youth/students)
• Municipalities
• Recycling operators
• Sponsor companies</td>
</tr>
<tr>
<td>Key Resources
• Smart-bin hardware & sensors
• AI verification software
• Recycling & user data</td>
<td>Channels
• Mobile app & QR codes
• Smart bins in public spaces
• Partner locations</td>
</tr>
<tr>
<td colspan="2">Cost Structure
• Smart-bin production & maintenance
• Software & data infrastructure
• Reward funding & operations</td>
<td colspan="3">Revenue Streams
• Government: pays a fixed amount to run and maintain the platform
• Private companies: commission on redeemed rewards (e.g. a user collects 1,000 points worth 10€ → we keep 10%)
• Future plans: a share from recycling (sale of collected materials) and in-app advertising</td>
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